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L'HOSPITAL'S THEOREM

Hardy, Amer. Math. Monthly 1951

L'Hospital's Rule (version)

Suppose f and g have continuous first derivatives with $g'(x) \neq 0$,

$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} g(x) = 0$, and

$$\lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = L.$$

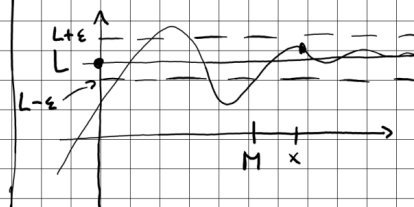
$$\text{Then } \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L.$$

Recall: We say $\lim_{x \rightarrow \infty} f(x) = L$

if for every $\varepsilon > 0$ there is $M > 0$

such that for all $x > M$ we

$$\text{have } |f(x) - L| < \varepsilon$$



proof: wlog, assume $g'(x) > 0 \quad \forall x \in \mathbb{R}$. (or at least for large x)

Given $\varepsilon > 0$ find $M > 0$ so that

$$\left| \frac{f'(x)}{g'(x)} - L \right| < \varepsilon \quad \forall x > M$$

$$\Rightarrow |f'(x) - Lg'(x)| < \varepsilon \cdot g'(x)$$

Note: $\forall t > M$ we have

$$\int_t^\infty |f'(x) - Lg'(x)| dx < \int_t^\infty \varepsilon g'(x) dx$$

$$\left| \int_t^\infty (f'(x) - Lg'(x)) dx \right| \rightarrow \left| f(x) - Lg(x) \right|_{x=t}^{x=\infty} < \varepsilon g(x) \Big|_{x=t}^{x=\infty}$$

$$\Rightarrow |f(t) - Lg(t)| < \varepsilon \cdot (-g(t))$$

$$\quad \quad \quad \parallel$$

$$\varepsilon |g(t)|$$

$$\Rightarrow \left| \frac{f(t)}{g(t)} - L \right| < \varepsilon.$$

note: $g'(x) > 0, g(x) \xrightarrow{x \rightarrow 0} 0$

$$\Rightarrow g(t) < 0$$

$$\Rightarrow -g(t) > 0$$

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